**SHETH L.U.J AND SIR M.V COLLEGE**
**SUBJECT : SCALA FOR DATA SCIENCE**

**MODULE 1**
**PRACTICAL NO - 2, 3, 4, 5**

**Aim :**
2 Practical Calculate mean, median, and mode of a list of numbers. Implement basic statistical calculations using Scala collections.
3. Generate a random dataset of 10 numbers and calculate its variance and standard deviation.
4. Create a dense vector using Breeze and calculate its sum, mean, and dot product with another vector.
5. Generate a random matrix using Breeze and compute its transpose and determinant.

**Practical 2**
**CODE :**
```
object Question1 extends App {

  val ticketsSold = List(
    42, 58, 37, 65, 31,
    72, 54, 49, 68, 45,
    81, 76, 52, 63, 47,
    70, 59, 42, 78, 55
  )

  // Mean
  val mean = ticketsSold.sum.toDouble / ticketsSold.length

  // Median
  val sortedList = ticketsSold.sorted
  val n = sortedList.length

  val median: Double =
    if (n % 2 == 0)
      (sortedList(n / 2 - 1) + sortedList(n / 2)) / 2.0
    else
      sortedList(n / 2).toDouble

  // Mode
  val mode = ticketsSold
    .groupBy(identity)
    .maxBy(_._2.size)
    ._1

  // Display Results
  println("========== Multiplex Cinema ==========")
  println("Movie Ticket Sales in 20 Days:")
  println(ticketsSold.mkString(", "))

  println("\nStatistical Calculations")
  println("------------------------")
```

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```
  println(f"Mean   : $mean%.2f")
  println(f"Median : $median%.2f")
  println(s"Mode   : $mode")
}
```

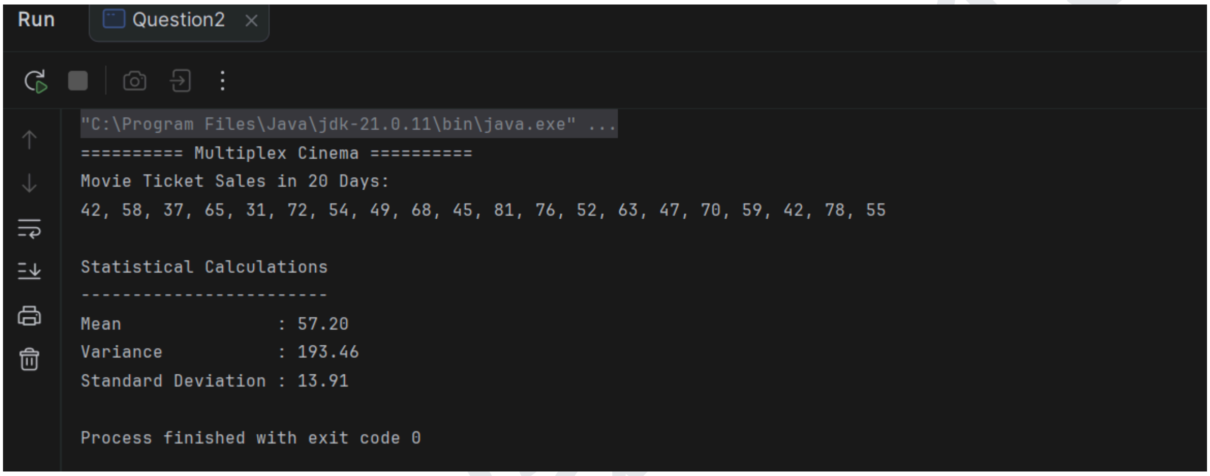
**OUTPUT :**


**Practical 3**
**CODE :**

```
import scala.math.sqrt

object Question2 extends App {

  val ticketsSold = List(
    42, 58, 37, 65, 31,
    72, 54, 49, 68, 45,
    81, 76, 52, 63, 47,
    70, 59, 42, 78, 55
  )

  // Mean
  val mean = ticketsSold.sum.toDouble / ticketsSold.length

  // Variance
  val variance = ticketsSold
    .map(x => math.pow(x - mean, 2))
    .sum / ticketsSold.length

  // Standard Deviation
  val standardDeviation = sqrt(variance)
```

```
  // Display Results
  println("========== Multiplex Cinema ==========")
  println("Movie Ticket Sales in 20 Days:")
  println(ticketsSold.mkString(", "))

  println("\nStatistical Calculations")
  println("------------------------")
  println(f"Mean               : $mean%.2f")
  println(f"Variance           : $variance%.2f")
  println(f"Standard Deviation : $standardDeviation%.2f")
}
```

**OUTPUT :**



```
Run   Question2
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
========== Multiplex Cinema ==========
Movie Ticket Sales in 20 Days:
42, 58, 37, 65, 31, 72, 54, 49, 68, 45, 81, 76, 52, 63, 47, 70, 59, 42, 78, 55

Statistical Calculations
------------------------
Mean               : 57.20
Variance           : 193.46
Standard Deviation : 13.91

Process finished with exit code 0
```

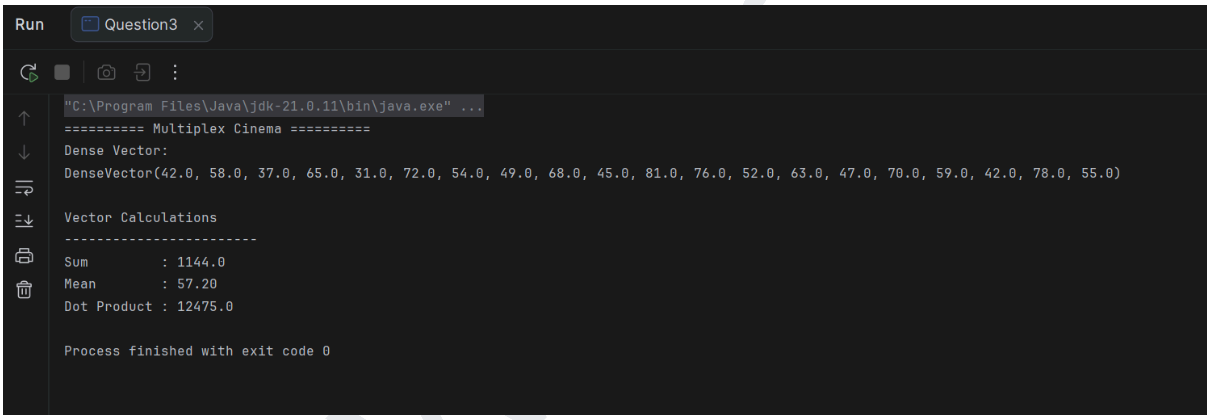
**Practical 4**

**CODE :**

```
import breeze.linalg._
object Question3 extends App {
  val ticketsSold = DenseVector(
    42.0, 58.0, 37.0, 65.0, 31.0,
    72.0, 54.0, 49.0, 68.0, 45.0,
    81.0, 76.0, 52.0, 63.0, 47.0,
    70.0, 59.0, 42.0, 78.0, 55.0
  )

  // Second vector (same size) for dot product
  val weights = DenseVector(
    1.0, 2.0, 3.0, 4.0, 5.0,
    6.0, 7.0, 8.0, 9.0, 10.0,
    11.0, 12.0, 13.0, 14.0, 15.0,
    16.0, 17.0, 18.0, 19.0, 20.0
  )

  // Calculations
  val totalSum = sum(ticketsSold)
```

```
  val mean = totalSum / ticketsSold.length
  val dotProduct = ticketsSold dot weights

  // Display Results
  println("========== Multiplex Cinema ==========")
  println("Dense Vector:")
  println(ticketsSold)

  println("\nVector Calculations")
  println("-----------------------")
  println(s"Sum         : $totalSum")
  println(f"Mean        : $mean%.2f")
  println(s"Dot Product : $dotProduct")

}
```

**OUTPUT :**



```
Run   Question3
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
========== Multiplex Cinema ==========
Dense Vector:
DenseVector(42.0, 58.0, 37.0, 65.0, 31.0, 72.0, 54.0, 49.0, 68.0, 45.0, 81.0, 76.0, 52.0, 63.0, 47.0, 70.0, 59.0, 42.0, 78.0, 55.0)

Vector Calculations
-----------------------
Sum         : 1144.0
Mean        : 57.20
Dot Product : 12475.0

Process finished with exit code 0
```

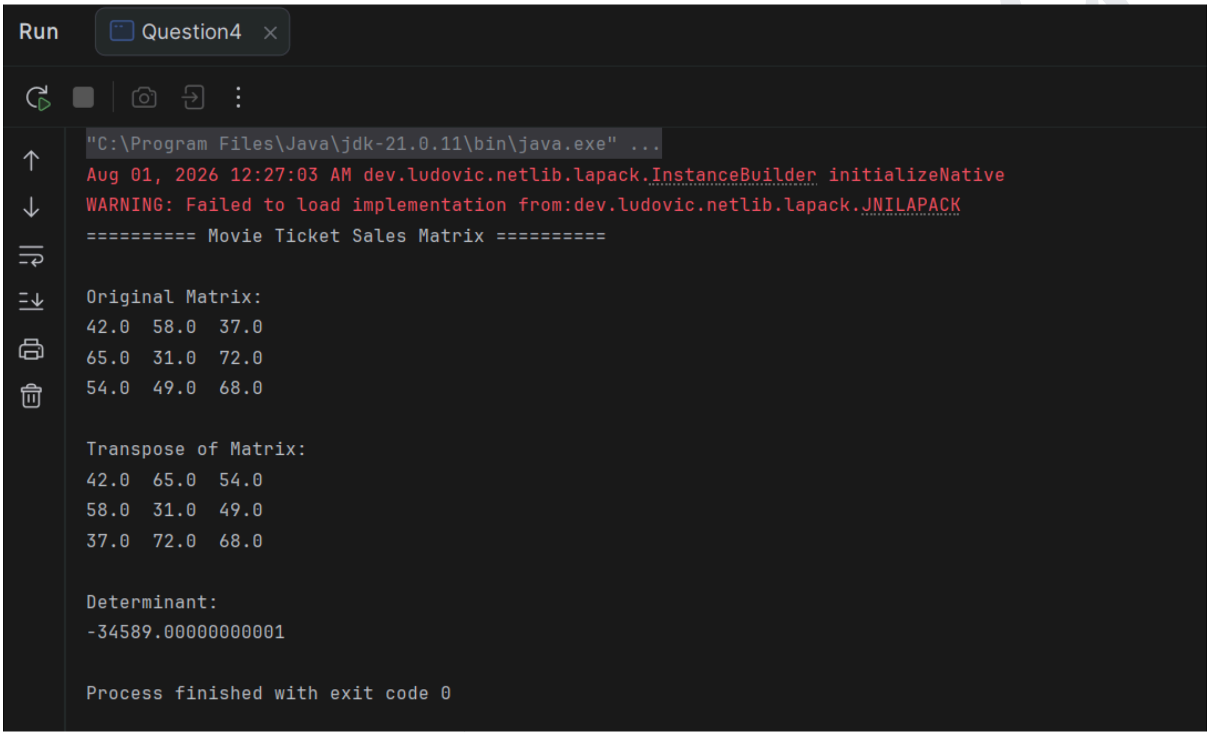
**Practical 5**

**CODE :**

```
import breeze.linalg._

object Question4 extends App {

  val matrix = DenseMatrix(
    (42.0, 58.0, 37.0),
    (65.0, 31.0, 72.0),
    (54.0, 49.0, 68.0)
  )

  val transposeMatrix = matrix.t
  val determinant = det(matrix)

  println("========== Movie Ticket Sales Matrix ==========")

  println("\nOriginal Matrix:")
```

```
  println(matrix)

  println("\nTranspose of Matrix:")
  println(transposeMatrix)

  println("\nDeterminant:")
  println(determinant)
}
```

**OUTPUT :**



```
Run   Question4
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
Aug 01, 2026 12:27:03 AM dev.ludovic.netlib.lapack.InstanceBuilder initializeNative
WARNING: Failed to load implementation from:dev.ludovic.netlib.lapack.JNILAPACK
========== Movie Ticket Sales Matrix ==========

Original Matrix:
42.0  58.0  37.0
65.0  31.0  72.0
54.0  49.0  68.0

Transpose of Matrix:
42.0  65.0  54.0
58.0  31.0  49.0
37.0  72.0  68.0

Determinant:
-34589.00000000001

Process finished with exit code 0
```

**Combined Practical 2,3,4,5 (Random Function Dataset)**
**CODE :**

```
import scala.util.Random
import scala.math.sqrt
import breeze.linalg._

object CombinedPractical extends App {
  val random = new Random()

  val data = List.fill(20)(random.nextInt(81) + 20)

  println("========================================")
  println(" RANDOM DATASET")
  println("========================================")
  println(data.mkString(", "))
```

```
val mean = data.sum.toDouble / data.length

val sortedData = data.sorted
val n = sortedData.length

val median: Double =
  if (n % 2 == 0)
    (sortedData(n / 2 - 1) + sortedData(n / 2)) / 2.0
  else
    sortedData(n / 2).toDouble

val mode = data
  .groupBy(identity)
  .maxBy(_._2.size)
  ._1

println("\n========== QUESTION 1 ==========")
println(f"Mean   : $mean%.2f")
println(f"Median : $median%.2f")
println(s"Mode   : $mode")

val variance = data
  .map(x => math.pow(x - mean, 2))
  .sum / data.length

val standardDeviation = sqrt(variance)

println("\n========== QUESTION 2 ==========")
println(f"Variance           : $variance%.2f")
println(f"Standard Deviation : $standardDeviation%.2f")

 val vector = DenseVector(data.map(_.toDouble).toArray)

val weights = DenseVector(
  (1 to 20).map(_.toDouble).toArray
)

val vectorSum = sum(vector)
val vectorMean = vectorSum / vector.length
val dotProduct = vector dot weights

println("\n========== QUESTION 3 ==========")
println("Dense Vector:")
println(vector)

println(f"\nSum         : $vectorSum%.2f")
println(f"Mean        : $vectorMean%.2f")
```

```
  println(f"Dot Product : $dotProduct%.2f")

   val matrix = DenseMatrix(
    (data(0).toDouble, data(1).toDouble, data(2).toDouble),
    (data(3).toDouble, data(4).toDouble, data(5).toDouble),
    (data(6).toDouble, data(7).toDouble, data(8).toDouble)
  )

  val transposeMatrix = matrix.t
  val determinant = det(matrix)

  println("\n========== QUESTION 4 ==========")

  println("\nOriginal Matrix:")
  println(matrix)

  println("\nTranspose Matrix:")
  println(transposeMatrix)

  println(f"\nDeterminant : $determinant%.2f")
}
```

**OUTPUT :**

```
Run   CombinedPractical

"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
========================================
 RANDOM DATASET
========================================
79, 48, 27, 52, 62, 87, 64, 68, 43, 49, 55, 83, 36, 84, 91, 83, 34, 81, 93, 39

========== QUESTION 1 ==========
Mean   : 62.90
Median : 63.00
Mode   : 83

========== QUESTION 2 ==========
Variance           : 427.79
Standard Deviation : 20.68

========== QUESTION 3 ==========
Dense Vector:
DenseVector(79.0, 48.0, 27.0, 52.0, 62.0, 87.0, 64.0, 68.0, 43.0, 49.0, 55.0, 83.0, 36.0, 84.0, 91.0, 83.0, 34.0, 81.0, 93.0, 39.0)

Sum         : 1258.00
Mean        : 62.90
Dot Product : 13686.00
Aug 01, 2026 12:30:53 AM dev.ludovic.netlib.lapack.InstanceBuilder initializeNative
WARNING: Failed to load implementation from:dev.ludovic.netlib.lapack.JNILAPACK

========== QUESTION 4 ==========

Original Matrix:
79.0  48.0  27.0
52.0  62.0  87.0
64.0  68.0  43.0
```

```
========== QUESTION 4 ==========

Original Matrix:
79.0  48.0  27.0
52.0  62.0  87.0
64.0  68.0  43.0

Transpose Matrix:
79.0  52.0  64.0
48.0  62.0  68.0
27.0  87.0  43.0

Determinant : -108478.00

Process finished with exit code 0
```